DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 12 OF 12

One-minute Concept Animation

Storyboard and supporting documentation for the 60-second film.

CONTENTS

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	**github.com/wasim437/Al_SAFA**
Live portal	**wasim437.github.io/Al_SAFA/**
Produced by	**tools/sync_film.py** · **tests/test_film.js**
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Al Safa 2 Park — Falaj Al Safa

Mohamed Wasim · Individual Applicant

Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

Concept Animation — Storyboard

Sixty seconds, drawn from the project's own analysis

The film is not an illustration of the design; it is rendered from the design. Its geometry comes from the same arc definition as the masterplan and its sun from the same 8,760-hour solar model, so it cannot drift away from the drawings the way an animation produced separately would.

Upload slot 12 — One-minute Concept Animation — Storyboard
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Structure — five scenes, sixty seconds

Time	Scene	What it shows
0–7 s	**The site**	Al Safa 2 as it is today: flat, exposed, and empty at midday
7–18 s	**The heat**	The reason it is empty — the modelled heat index across a summer day, and the hours that are lost to it
18–38 s	**The crescent**	The arc builds and plants itself while a full computed day passes over it. 131 trees, the falaj, the radial alleys and the rooms appear in the order they are reached
38–48 s	**Beneath it**	Eye level on the walk: the dappled soffit, the louvre cutting the low sun, the channel at the edge
48–60 s	**The proof**	The measured result — comfortable hours rising from 44.5% to 64.6% — as the park fills at dusk

Scene boundaries are asserted by tests/test_film.js, which renders every frame and fails on any gap, overlap or invalid value.

2. Why it can be trusted against the drawings

- The arc is the same 141 m radius as the masterplan, regenerated by tools/sync_film.py from src/plan.py.
- The sun positions are the same NREL-computed values used by the shade model — the shadows move correctly for the site's latitude and the time of year being shown.
- The tree count, the 0.9 m channel and the room layout are read from the same geometry as every drawing in this submission.
- Every frame is rendered by an automated test that fails on any NaN or undefined value, so a geometry change cannot break the film quietly.

3. How to produce the video file

- Open `concept_film.html` full-screen in a browser.
- Start a screen recording (Win + Alt + R on Windows, Shift + Cmd + 5 on macOS).
- Press Restart and let all sixty seconds run.
- Stop, trim the ends, and add a voiceover if wanted.

The film shows the **analysed** scheme exactly: the same arc, the same trees, the same solar model. It is therefore consistent with every number quoted elsewhere in this submission.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.